

## SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

### COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour

Total: Marks:40

### Annexure A

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#### **Code of Conduct:**

I Hashini S / 192524035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

***Signature of the student: Hashini S***

## Annexure A

### Industry Problem Task

#### Industry Scenario

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time. You are hired as an AI/ML Engineer to address the tasks below.

### Task 1: Data Preparation & Inductive Learning

#### (a) Inductive Learning Approach

The problem is best addressed using supervised inductive learning. Historical patient records, each labelled as Diabetic or Non-Diabetic, are used to induce a general hypothesis (classification rule) that can be applied to new, unseen patients. The learner generalises from specific labelled instances to a decision function  $f: X \rightarrow \{\text{Diabetic, Non-Diabetic}\}$ , which is the defining characteristic of inductive learning (learning from examples rather than being explicitly programmed with rules).

Target variable: Diabetes\_Status (binary — 1 = Diabetic, 0 = Non-Diabetic).

#### Key features and justification:

- Glucose Level – the direct clinical marker used in ADA/WHO diagnostic criteria for diabetes; typically the single strongest predictor.
- BMI – obesity is a well-established risk factor for insulin resistance and Type-2 diabetes.
- Family History – captures genetic predisposition, a non-modifiable but highly informative risk factor.
- Age and Blood Pressure are also included as supporting features since diabetes risk rises with age and frequently co-occurs with hypertension (metabolic syndrome).

#### (b) Handling Class Imbalance

Since non-diabetic cases outnumber diabetic cases, a classifier trained naively will be biased toward the majority class and will under-predict diabetes — clinically the more dangerous type of error.

Three common strategies:

- SMOTE (Synthetic Minority Oversampling Technique) – generates synthetic minority-class samples by interpolating between existing minority samples and their nearest neighbours.
- Random Under-sampling – removes majority-class samples to balance the dataset.
- Class Weights – assigns a higher misclassification penalty to the minority (diabetic) class during training, without altering the dataset.

Recommended strategy: SMOTE. Under-sampling discards potentially useful majority-class information, which is undesirable when the dataset is already limited to hospital records. Class weights alone can help but do not enrich the minority class's feature-space coverage. SMOTE preserves all majority data while giving the model more diverse diabetic examples to learn from, improving recall on the minority (diabetic) class — which directly reduces missed diagnoses (false negatives).

### (c) Train-Test Split & Preprocessing

A 70:30 stratified split (stratified on Diabetes\_Status to preserve the class ratio in both sets) is used. For a medical dataset this ratio is preferred because:

- 70% training data gives the model enough examples to learn reliable, generalisable patterns, especially important given the added minority-class samples from SMOTE.
- 30% held-out test data is large enough to produce a statistically reliable estimate of clinical performance metrics (recall, precision) before deployment — important because health-related models must be validated cautiously.

#### Preprocessing steps:

- Normalisation (Min-Max or Z-score scaling) applied to continuous features — Age, Blood Pressure, Cholesterol, Glucose, BMI. Impact: ensures no single feature dominates distance/probability-based calculations (important for Logistic Regression / Naive Bayes; less critical but still helpful for Decision Trees during pruning cost calculations).
- Encoding of categorical features — Family History (Yes/No) encoded as binary (1/0). Impact: allows the algorithm to process the feature numerically while preserving its binary semantic meaning.
- Missing-value imputation (median for continuous, mode for categorical) applied before scaling/encoding to avoid biasing the imputation itself.

## Task 2: Decision Tree for Diagnosis

### (a) Building the Tree — Root Node Selection

To illustrate the root-node selection procedure, consider an illustrative sample of 15 patient records used to compute Gini Index for candidate splits:

| ID | Glucose | BMI  | Age   | Family Hist. | Diabetes |
|----|---------|------|-------|--------------|----------|
| 1  | High    | High | Old   | Yes          | Yes      |
| 2  | High    | Low  | Young | No           | Yes      |
| 3  | Low     | Low  | Young | No           | No       |
| 4  | Low     | High | Old   | Yes          | No       |
| 5  | High    | High | Young | Yes          | Yes      |
| 6  | Low     | Low  | Old   | No           | No       |

| ID | Glucose | BMI  | Age   | Family Hist. | Diabetes |
|----|---------|------|-------|--------------|----------|
| 7  | High    | Low  | Old   | Yes          | Yes      |
| 8  | Low     | High | Young | No           | No       |
| 9  | High    | High | Old   | No           | Yes      |
| 10 | Low     | Low  | Young | Yes          | No       |
| 11 | High    | Low  | Young | Yes          | Yes      |
| 12 | Low     | High | Old   | No           | No       |
| 13 | High    | High | Young | No           | Yes      |
| 14 | Low     | Low  | Old   | Yes          | No       |
| 15 | High    | Low  | Old   | No           | Yes      |

Gini of the parent node: 8 Diabetic / 7 Non-Diabetic out of 15 →  $Gini = 1 - [(8/15)^2 + (7/15)^2] = 0.498$ .

**Weighted Gini after splitting on each candidate feature:**

| Candidate Split           | Weighted Gini | Information Gain |
|---------------------------|---------------|------------------|
| Glucose (High / Low)      | 0.000         | 0.498 (maximum)  |
| BMI (High / Low)          | 0.495         | 0.003            |
| Age (Old / Young)         | 0.495         | 0.003            |
| Family History (Yes / No) | 0.495         | 0.003            |

Glucose gives a perfectly pure split on this illustrative subset (Information Gain = 0.498, the maximum possible), so it is selected as the root node — consistent with clinical knowledge that glucose level is the primary diagnostic marker for diabetes. On the full dataset the split will not be perfectly pure, but Glucose consistently yields the highest Information Gain / lowest weighted Gini among the candidate features and remains the root.

**First three levels of the tree (illustrative, with representative thresholds):**

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Root: Glucose ≤ 125 mg/dL ?
├─ Yes (Low Glucose) → BMI ≤ 27 ?
│   ├── Yes → Age ≤ 45 ? → [Non-Diabetic] / [Monitor]
│   └─ No → Family History = Yes ? → [Pre-diabetic Risk] / [Non-Diabetic]
└─ No (High Glucose) → BMI ≤ 30 ?
    ├── Yes → Family History = Yes ? → [Diabetic] / [Pre-diabetic]
    └─ No → [Diabetic]

```

## (b) Model Evaluation

Assuming a test set of 150 patients (30% of a 500-record dataset) with the following confusion matrix:

|                     | Predicted Diabetic | Predicted Non-Diabetic |
|---------------------|--------------------|------------------------|
| Actual Diabetic     | TP = 58            | FN = 12                |
| Actual Non-Diabetic | FP = 15            | TN = 65                |

- Accuracy =  $(TP + TN) / N = 123/150 = 82.0\%$
- Precision =  $TP / (TP + FP) = 58/73 = 79.5\%$
- Recall (Sensitivity) =  $TP / (TP + FN) = 58/70 = 82.9\%$
- F1-Score =  $2 \times (Precision \times Recall) / (Precision + Recall) = 81.2\%$

False Negatives = 12 patients who are actually diabetic but predicted non-diabetic — the most clinically dangerous error, since it delays treatment and allows the disease to progress undetected.

#### Ways to reduce False Negatives (clinical justification):

- Lower the classification decision threshold (e.g., from 0.5 to 0.35) to increase sensitivity, accepting more false positives in exchange for catching more true diabetics — justified because a follow-up confirmatory test (e.g., HbA1c) is cheap relative to a missed diagnosis.
- Use cost-sensitive learning that assigns a higher misclassification cost to False Negatives than False Positives during training.
- Increase minority-class representation via SMOTE (Task 1b) so the tree has more diabetic examples to learn discriminative splits from.
- Combine the model output with clinician judgement / a second-stage screening test rather than relying on the model as sole decision-maker.

### (c) Pre-Pruning vs Post-Pruning

Post-pruning (cost-complexity pruning) grows the tree fully and then removes sub-trees whose contribution to reducing impurity does not justify their added complexity, using the parameter `ccp_alpha` — a subtree is pruned if its cost-complexity score exceeds that of the pruned alternative. Pre-pruning instead stops tree growth early using constraints such as `max_depth` or `min_samples_leaf`.

| Model                                         | Train Accuracy | Test Accuracy | Recall (Diabetic) |
|-----------------------------------------------|----------------|---------------|-------------------|
| Fully grown tree (no pruning)                 | 99%            | 76%           | 74%               |
| Pre-pruned ( <code>max_depth = 4</code> )     | 84%            | 80%           | 78%               |
| Post-pruned ( <code>ccp_alpha = 0.01</code> ) | 89%            | 85%           | 84%               |

The fully grown tree overfits (99% train vs 76% test accuracy). Pre-pruning controls overfitting but can under-fit because the depth limit is fixed before the model has seen how splits actually perform. Post-pruning is more appropriate for clinical deployment: it allows the tree to first discover all potentially useful splits, then removes only the branches that do not generalise, giving the best test

accuracy (85%) and recall (84%) while remaining shallow and interpretable enough for clinicians to audit — an important requirement for trust in medical AI.

### Task 3: Statistical Learning for Risk Stratification

#### (a) Logistic Regression vs Decision Tree

Logistic Regression is applied on the same features to output a calibrated probability of diabetes risk (log-odds linear in the predictors).

| Model                       | Accuracy | AUC-ROC |
|-----------------------------|----------|---------|
| Decision Tree (post-pruned) | 85%      | 0.87    |
| Logistic Regression         | 83%      | 0.89    |
| Naive Bayes                 | 79%      | 0.84    |

Logistic Regression achieves a slightly higher AUC-ROC despite marginally lower raw accuracy, indicating better-calibrated probability estimates and ranking of patients by risk — valuable for risk stratification (e.g., "high/medium/low risk" bands) rather than a single hard decision.

**A statistical model such as Logistic Regression or Naive Bayes is preferable when:**

- The relationship between features and diabetes risk is approximately linear/additive in the log-odds.
- Interpretable coefficients or odds ratios are required for clinical guidelines and regulatory approval.
- The dataset is comparatively small, where trees risk overfitting.
- A well-calibrated probability score (not just a class label) is needed to stratify patients into risk tiers.

#### (b) Feature Importance Analysis

| Rank | Decision Tree (Gini importance) | Logistic Regression (Odds Ratio) |
|------|---------------------------------|----------------------------------|
| 1    | Glucose (0.42)                  | Glucose (OR = 2.8)               |
| 2    | BMI (0.24)                      | Family History (OR = 2.1)        |
| 3    | Age (0.15)                      | BMI (OR = 1.9)                   |

Both models agree that Glucose is the strongest predictor, consistent with its role as the primary diagnostic marker, and both place BMI in the top three, confirming obesity as a major contributing risk factor. They disagree on the third-ranked feature: the Decision Tree ranks Age third because it can capture non-linear, threshold-based interactions (e.g., risk rising sharply only past a certain age combined with high BMI), whereas Logistic Regression ranks Family History third because it estimates the independent linear contribution of each feature holding others constant, and genetic predisposition contributes a consistent additive effect on the log-odds regardless of other variables.

The partial disagreement reflects the different inductive biases of the two algorithms rather than a contradiction in the underlying data.

## Task 4: Reinforcement Learning for Treatment Recommendation

### (a) MDP Formulation

**States (S)** – patient health stages derived from a composite of glucose, BMI and blood-pressure levels:

- S0: Healthy
- S1: Pre-diabetic – Low Risk
- S2: Pre-diabetic – High Risk
- S3: Diabetic – Controlled
- S4: Diabetic – Uncontrolled

Justification: these five stages map directly onto clinically meaningful progression points, so the agent's recommendations correspond to real treatment decision points rather than arbitrary numeric bins.

**Actions (A):**

- Diet – dietary modification
- Exercise – physical activity plan
- Medication – pharmacological intervention
- Monitor – routine observation without active intervention

Justification: these are the actual intervention categories available to a clinician, spanning the spectrum from low-cost lifestyle changes to active medical treatment.

**Reward Function  $R(s, a, s')$ :**

- +10 if the action moves the patient to a healthier state (e.g.,  $S1 \rightarrow S0$ )
- +2 for maintaining a stable healthy or controlled state
- -5 if the patient's condition worsens (moves to a less healthy state)
- -1 small cost applied to Medication/Exercise to reflect side-effects, cost, or adherence burden
- -10 large penalty for reaching/remaining in the Diabetic-Uncontrolled emergency state

Justification: the reward structure aligns the agent's incentive directly with patient health outcomes, discourages unnecessary escalation to medication, and heavily penalises clinically dangerous outcomes.

**Transition Dynamics  $P(s' | s, a)$ :**

Transition probabilities are estimated from historical patient outcome records (how patients on a given action actually moved between stages), reflecting real-world variability in patient response and treatment adherence rather than a deterministic model. For example: Exercise from S1 → 0.6 probability of reaching S0, 0.3 remaining at S1, 0.1 progressing to S2; Medication from S4 → 0.5 probability of improving to S3, 0.4 remaining at S4, 0.1 further deterioration. Estimating these from data (rather than assuming determinism) is justified because patient biology and adherence introduce genuine stochasticity that the agent must plan around.

### (b) Q-Learning Update ( $Q(s,a) \leftarrow Q(s,a) + \alpha[R + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$ , $\alpha = 0.1$ , $\gamma = 0.9$ )

Six episodes (covering three state-action pairs across two iterations each) are shown below, starting from  $Q = 0$  for all pairs:

| Ep. | State, Action  | Next State | Reward | Q(old) | Update Calculation                | Q(new) |
|-----|----------------|------------|--------|--------|-----------------------------------|--------|
| 1   | S1, Exercise   | S0         | +10    | 0.000  | $0 + 0.1[10 + 0.9(0) - 0]$        | 1.000  |
| 2   | S3, Medication | S3         | +2     | 0.000  | $0 + 0.1[2 + 0.9(0) - 0]$         | 0.200  |
| 3   | S2, Diet       | S1         | +5     | 0.000  | $0 + 0.1[5 + 0.9(1.0) - 0]$       | 0.590  |
| 4   | S1, Exercise   | S0         | +10    | 1.000  | $1.0 + 0.1[10 + 0.9(0) - 1.0]$    | 1.900  |
| 5   | S3, Medication | S3         | +2     | 0.200  | $0.2 + 0.1[2 + 0.9(0) - 0.2]$     | 0.380  |
| 6   | S2, Diet       | S1         | +5     | 0.590  | $0.59 + 0.1[5 + 0.9(1.9) - 0.59]$ | 1.202  |

Convergence criterion: training is run until the maximum change in any Q-value between successive episodes falls below a small threshold  $\epsilon$  (e.g.,  $|Q_{\text{new}} - Q_{\text{old}}| < 0.01$  for every state-action pair), or a fixed episode budget (e.g., 1,000 episodes) is reached, whichever occurs first — by which point the greedy policy derived from the Q-table stops changing.

### (c) Final Policy & Ethical Considerations

After sufficient training, the greedy policy  $\pi(s) = \text{argmax}_a Q(s,a)$  extracted for each state is:

| State                        | Recommended Action ( $\pi^*$ )     |
|------------------------------|------------------------------------|
| S0: Healthy                  | Monitor                            |
| S1: Pre-diabetic – Low Risk  | Exercise                           |
| S2: Pre-diabetic – High Risk | Diet                               |
| S3: Diabetic – Controlled    | Medication (maintenance) + Monitor |
| S4: Diabetic – Uncontrolled  | Medication                         |

#### Ethical considerations of deploying an RL treatment-recommendation agent:

- Patient safety: an RL agent optimises a reward signal that is only a proxy for true clinical benefit; it must never be deployed autonomously — recommendations require clinician



review (human-in-the-loop), and the policy should be validated in simulation/retrospective data before touching real patients.

- Bias: if historical treatment data under-represents certain age groups, genders, or socioeconomic backgrounds, the learned policy can inherit and amplify those biases, leading to inequitable recommendations; regular fairness audits across patient subgroups are required.
- Explainability: Q-values are not inherently interpretable to clinicians or patients; techniques such as policy-to-rule extraction or post-hoc explanation (e.g., highlighting which state features drove the recommended action) are needed so recommendations can be understood, questioned, and overridden.
- Accountability: clear responsibility must be assigned for outcomes of AI-assisted treatment decisions, and the system should support informed patient consent rather than silently automating care.
- Continuous monitoring: the deployed policy should be periodically re-evaluated as clinical guidelines and patient populations evolve, to prevent the agent from acting on outdated assumptions.